if true, would happen very slowly) and the possible transmutations and decays of various combinations of quarks, in violation of certain long-established properties of point-particle quantum field theory.20 These kinds of processes are especially interesting because their absence from conventional theory makes them sensitive signals of physics that cannot be accounted for without invoking new theoretical principles. If observed, any one of these processes would provide fertile ground for string theory to offer an explanation. Third, for certain choices of the Calabi-Yau shape there are particular patterns of string vibration that can effectively contribute new, tiny, long-range force fields. Should the effects of any such new forces be discovered, they might well reflect some of the new physics of string theory. Fourth, as we note in the next chapter, astronomers have collected evidence that our galaxy and possibly the whole of the universe is immersed in a bath of dark matter, the identity of which has yet to be determined. Through its many possible patterns of resonant vibration, string theory suggests a number of candidates for the dark matter; the verdict on these candidates must await future experimental results establishing the detailed properties of the dark matter.

And finally, a fifth possible means of connecting string theory to observations involves the cosmological constant—remember, as discussed in Chapter 3, this is the modification Einstein temporarily imposed on his original equations of general relativity to ensure a static universe. Although the subsequent discovery that the universe is expanding led Einstein to retract the modification, physicists have since realized that there is no explanation for why the cosmological constant should be zero. In fact, the cosmological constant can be interpreted as a kind of overall energy stored in the vacuum of space, and hence its value should be theoretically calculable and experimentally measurable. But, to date, such calculations and measurements lead to a colossal mismatch: Observations show that the cosmological constant is either zero (as Einstein ultimately suggested) or quite small; calculations indicate that quantum-mechanical fluctuations in the vacuum of empty space tend to generate a nonzero cosmological constant whose value is some 120 orders of magnitude (a 1 followed by 120 zeros) larger than experiment allows! This presents a wonderful challenge and opportunity for string theorists: Can